

AI Impact Analysis Based on 200K AI/ML News

Topic → Industry mapping · Entity extraction · Sentiment model (trained) · Over-time insights

ADSP 32018 IP01 Next-Gen NLP: Transformers, LLMs and AgenticAI in Practice

WINTER 2026 - Final Project

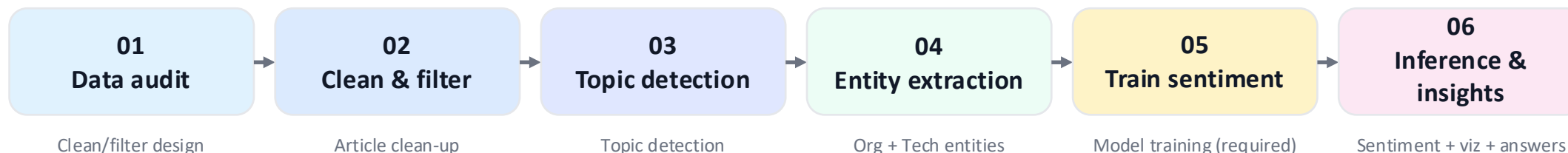
Deliverable: business-facing insights answering 3 questions

- Which industries & companies are most likely to be impacted by AI in the future?
- How will they be impacted (positive/negative/unclear) and by what means / technologies?
- What factors are associated with successful vs unsuccessful AI adoption?

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Date: 03/10/2026 (MM/DD/YYYY)

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How components connect (why this is business-facing)

- Filtering focuses analysis on evidence-rich AI impact paragraphs (reduces web-crawl noise).
 - Define AI trigger patterns and extract evidence-rich paragraphs
 - Score each document using high-signal and supporting AI terms
- Topics are mapped into an industry taxonomy → scalable “who is impacted” view.
 - Fit BERTopic on the prepared corpus
 - Map BERTopic topics to an industry taxonomy
 - The mapping is based on semantic similarity between topic embeddings and manually crafted industry anchor texts.
- Entities (companies + technologies) provide drill-down and the “by what means” lens.
 - Leverage distilbert-NER to extract Organization Entities
 - Extract technology entities by Rule-Based Phrase Matching
- A custom-trained sentiment model enables consistent labeling at scale (millions of contexts).
 - Construct sentence-based context windows around organization mentions
 - Create weak labels with an LLM (GPT 5-nano)
 - Fine-Tune roberta-base with Class-Weighted Loss
- Aggregations over (industry × company × technology × time) produce the final insights.
 - Inference the whole corpus and produce plots

Articles (raw)

199,989

Date range

2022-01 → 2026-02

Schema

url · date · language · title · text

Figure: Monthly article volume (raw)

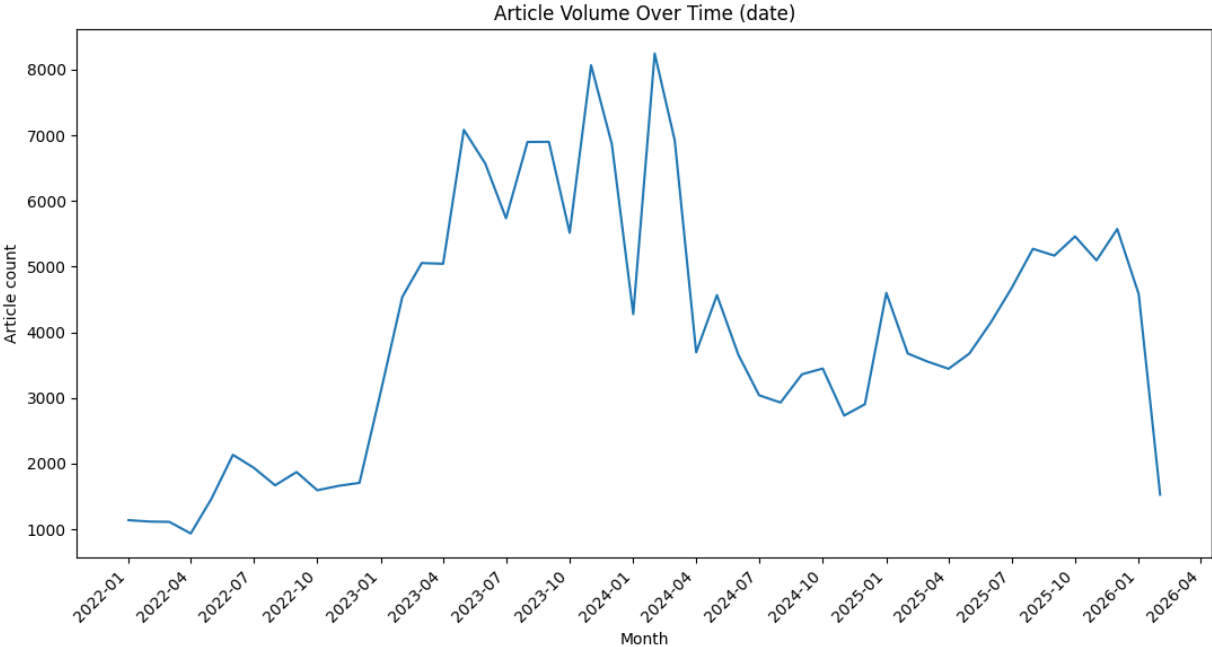
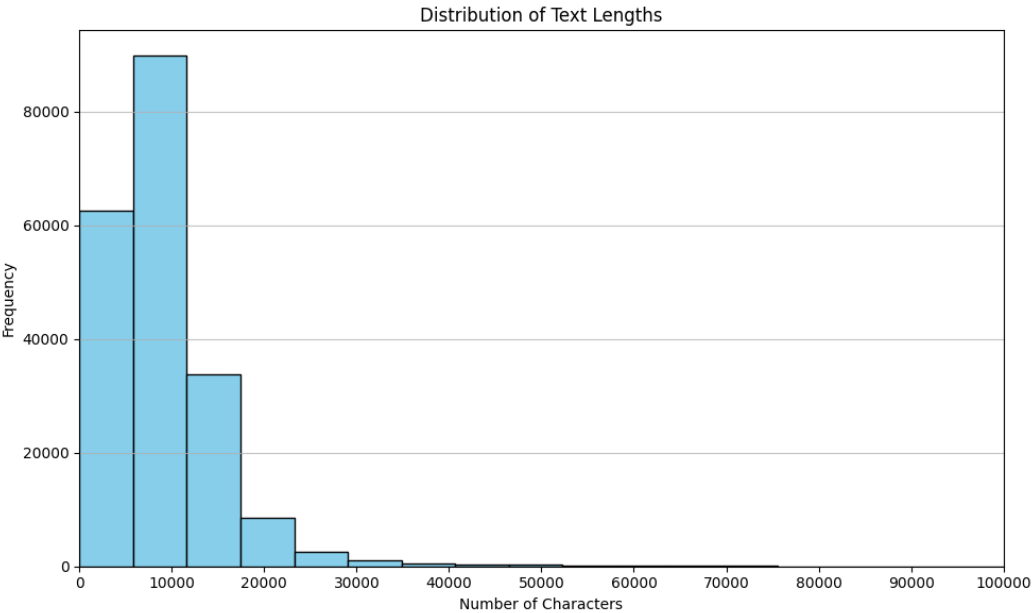
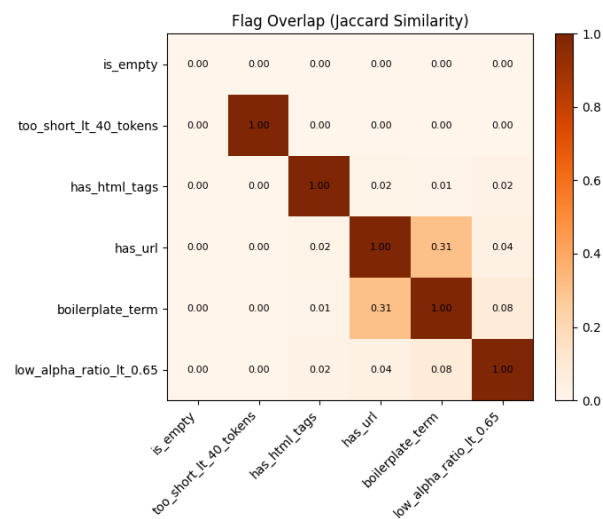
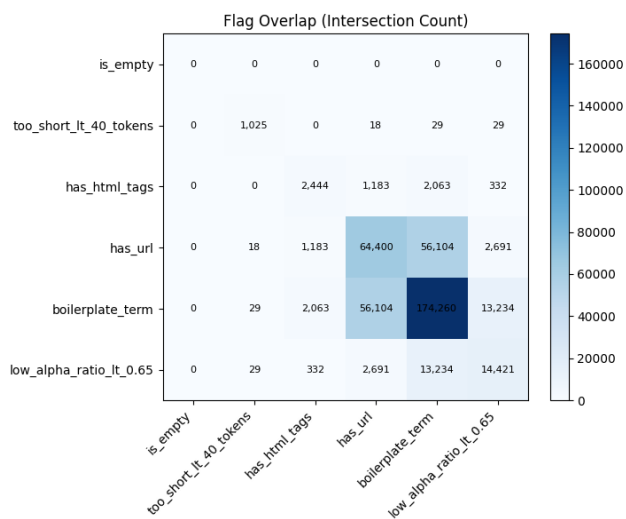
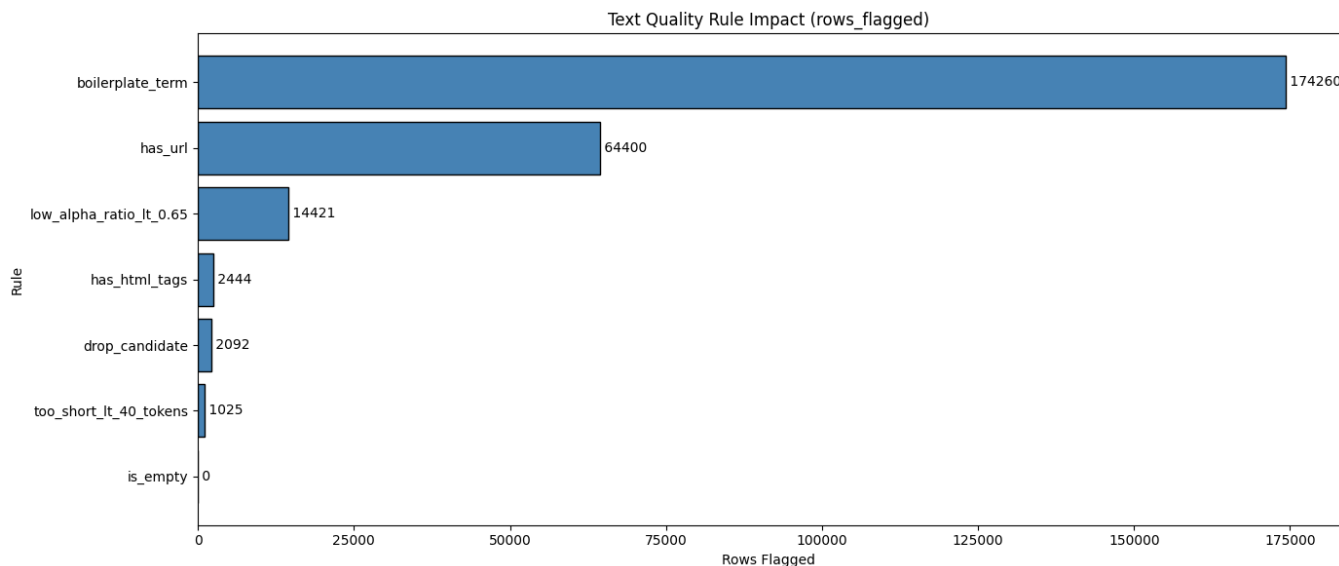


Figure: Text length distribution (raw)



Heuristic flags on raw article text



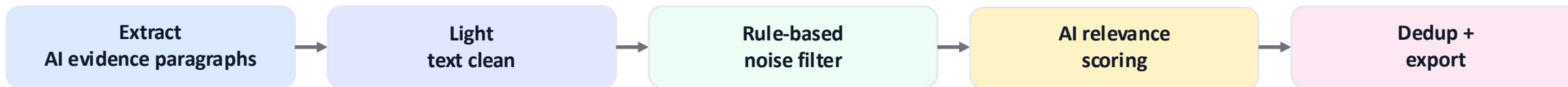
Observed artifacts (examples)

- Boilerplate (privacy policy, subscribe banners, navigation menus)
- Copy/paste link lists and “related stories” blocks
- HTML / markup remnants and low alphabetic ratio noise
- Extremely short or non-informative pages (image galleries, stubs)

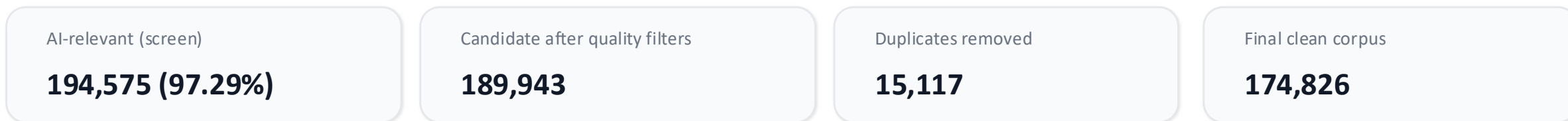
Data-driven filter impact (from audit)

- Drop boilerplate-heavy pages: ~2,063 rows ($\approx 1.03\%$)
 - Deduplicate normalized text: ~511 rows ($\approx 0.26\%$)
 - Drop short/low-info rows: ~29 rows ($\approx 0.01\%$)
- Audit suggests targeted rules can remove noise without shrinking corpus drastically

Goal: produce a reusable, high-signal AI impact corpus for downstream modeling



Key dataset reductions (transparent counts)



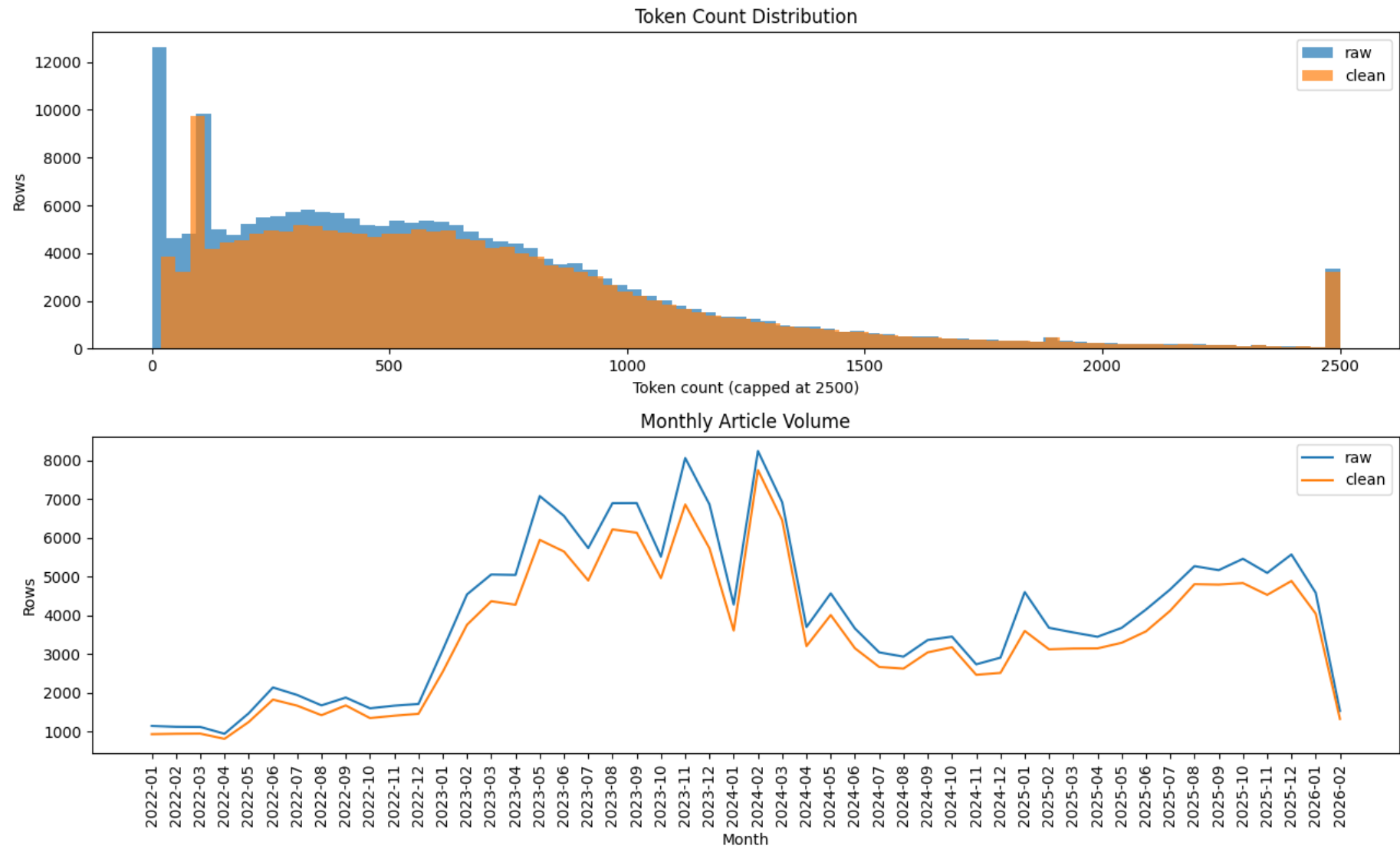
Why this matters?

- Downstream modeling (topics/entities/sentiment) is done on the cleaned, evidence-rich text — not raw webpage artifacts.
- Filtering is rule-driven and auditable (counts reported), rather than “black-box” removal.
- The clean corpus is exported as a reusable artifact for reproducibility.

How did I filter the corpus?

- Extract AI evidence paragraphs
 - split each article into paragraphs and kept only paragraphs containing explicit AI/ML trigger terms such as *AI*, *machine learning*, *LLM*, *generative AI*, *ChatGPT*, and *OpenAI*.
 - This reduced webpage clutter and turned each article into a more focused, AI-centered text unit.
- Apply light text cleaning
 - lightly normalized titles and extracted text by removing HTML tags, URLs, repeated separators, and extra whitespace.
 - created cleaned text fields and basic diagnostics such as **character count**, **token count**, and **alphabetic-character ratio**.
- Apply staged filtering and deduplicate

Token-length and time trends: cleaned corpus preserves global structure while reducing noise



Approach: BERTopic on cleaned corpus, then map topic clusters to business-friendly industries

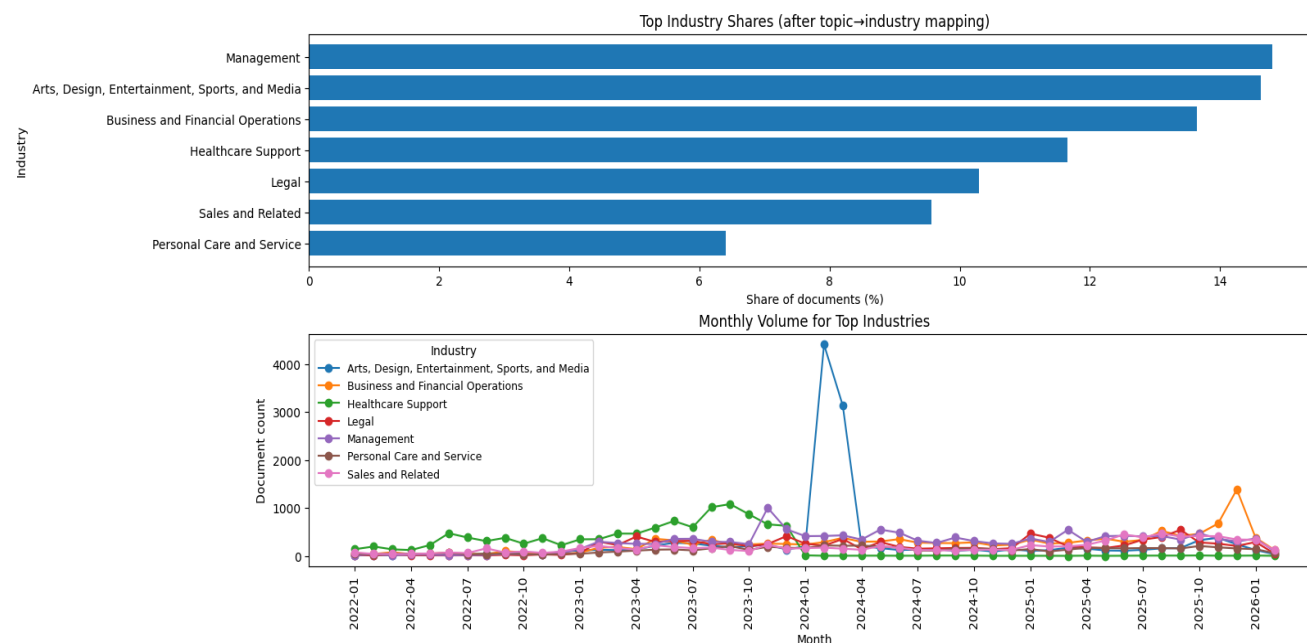
Modeling steps

- Prepare modeling text from evidence-rich paragraphs
- Embed documents (sentence embeddings)
- Cluster embeddings (HDBSCAN) + topic representation (c-TF-IDF)
- Assign topic_id per document (incl. -1 outliers)
- Map topic→industry taxonomy for interpretation and aggregation

Outputs used downstream

- Document-level topic assignments
- Industry-level prevalence (% share) and monthly volume
- Industry tags joined with entity + sentiment results

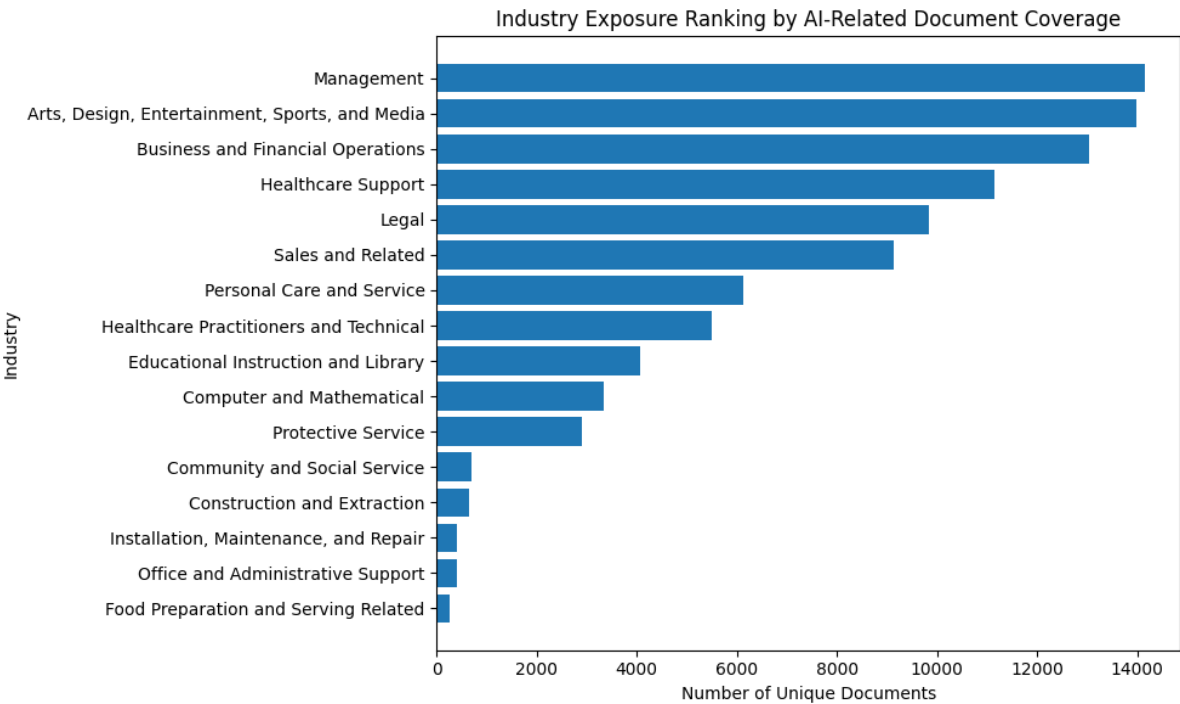
Example output: industry shares + monthly volume



Q1: Which industries are most likely to be impacted?

Proxy used here: “AI impact exposure” = volume of AI-related coverage mapped to each industry

- The heaviest AI-impact coverage concentrates in **Management, Arts/Media, Business & Financial Operations, Healthcare Support, Legal, and Sales**, suggesting that AI discussion is not limited to “tech-native” domains but is spreading into a broad set of knowledge and service occupations.
- These industries appear most frequently in the cleaned news corpus, meaning they are the places where firms, workers, and workflows are already being actively discussed in relation to AI adoption.
- By mapping BERTopic clusters into a business-friendly industry taxonomy, the project converts fragmented article-level discussions into an interpretable **industry impact map**, making it easier to compare where AI attention is most concentrated.



Left: unique-document exposure · Right: composite impact (volume-weighted)

Two-track extraction: pretrained NER for organizations + rule-based phrase matching for technologies

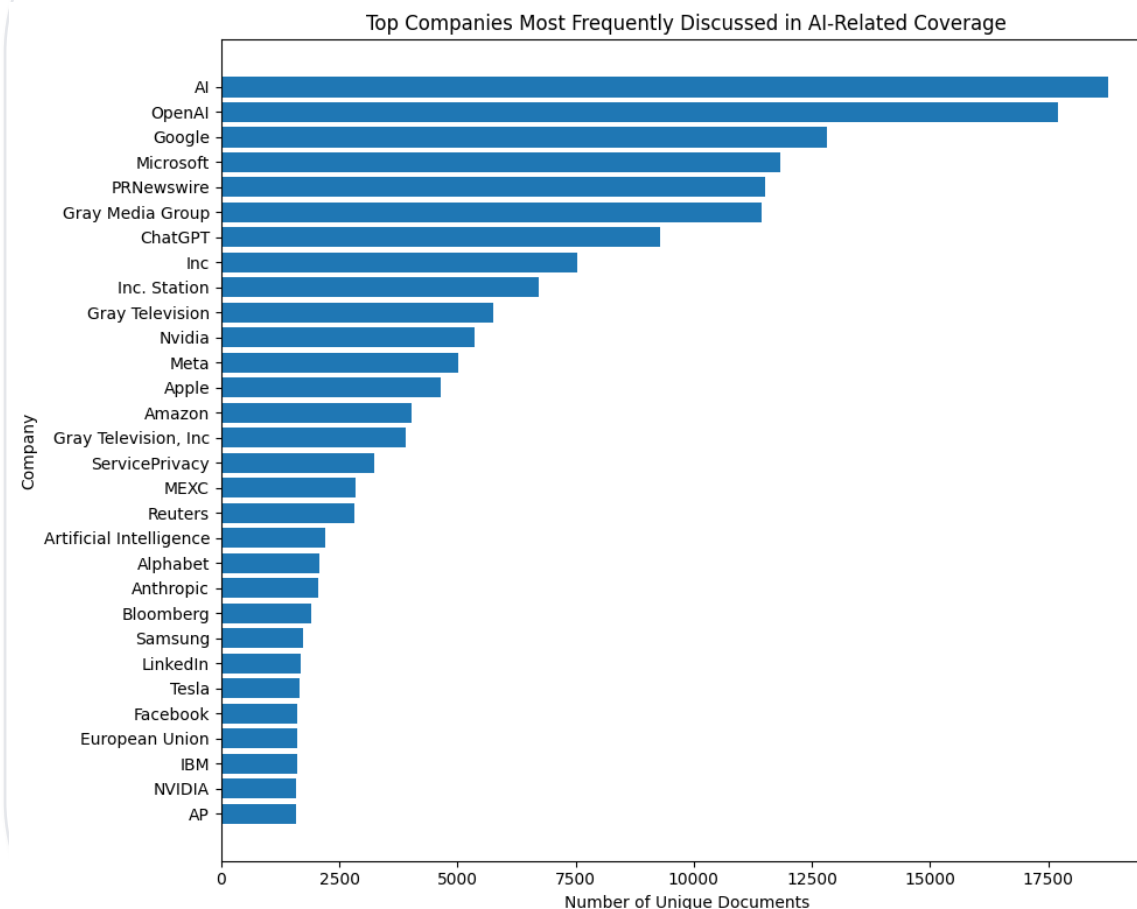
Extraction & normalization

- Org NER: detect organization mentions (e.g., OpenAI, Microsoft, Nvidia)
 - Leveraged distilbert-NER to extract ORG entities
- Tech phrases: curated lexicon + phrase matcher (e.g., process automation, AI copilots)
- Standardize variants and join back to document metadata (date, topic/industry)
- Aggregate by industry/company/technology/time for insights

Why entities are critical

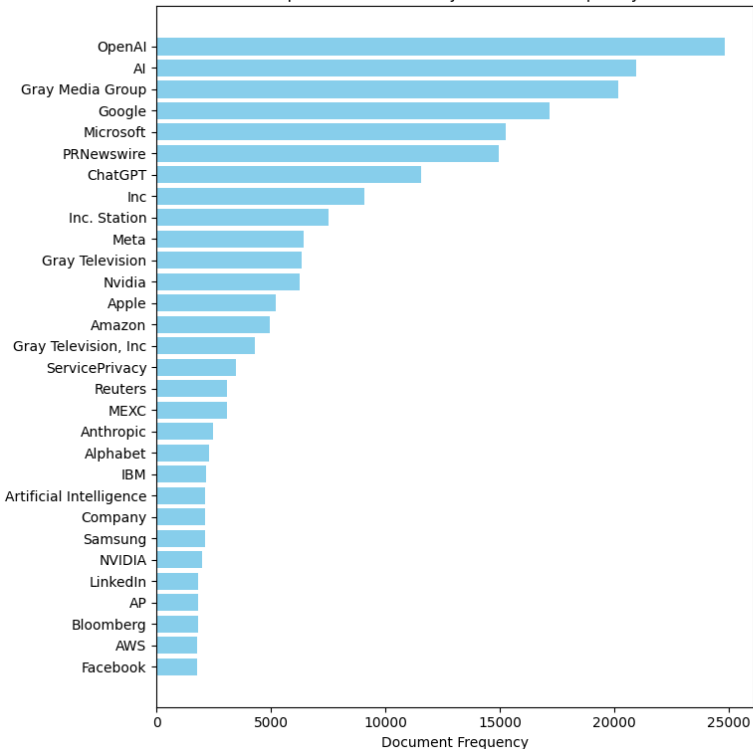
- Topics explain “industry-level narrative”.
- Entities provide the drill-down: “which companies” + “by what means”.
- Entity-centered contexts become the unit for sentiment training + inference.

Example output: top companies in AI-related coverage

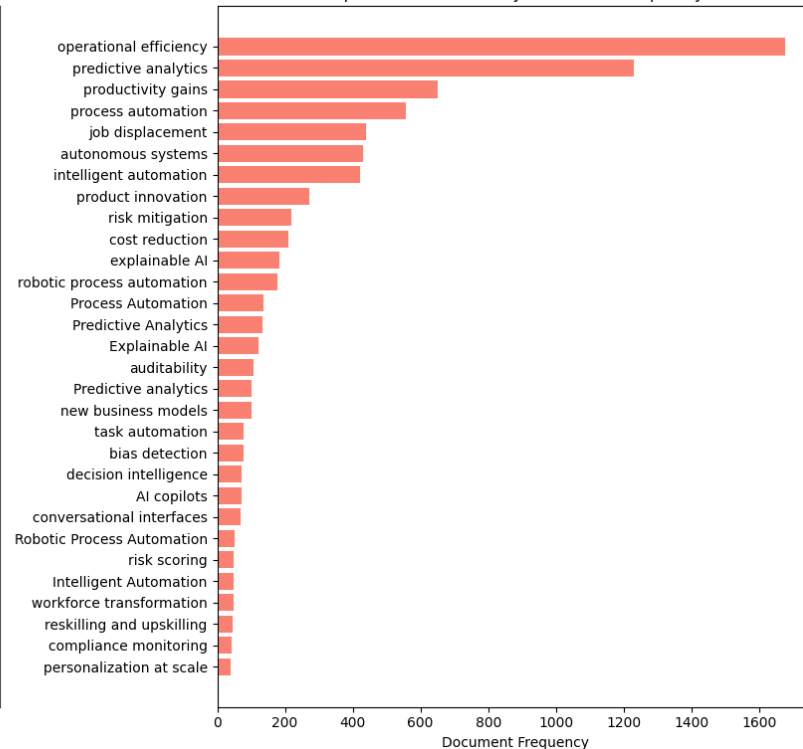


Top companies/organizations in AI impact coverage (unique documents)

Top 30 ORG Entities by Document Frequency



Top 30 TECH Entities by Document Frequency



Interpretation notes

- Counts reflect how often entities appear in AI impact contexts, not company size.
- Some “entities” are publishers or generic terms (e.g., PRNewswire) → limitation to improve with stronger canonicalization.
- We use entity-level sentiment (next slides) to distinguish “hyped” vs “criticized” coverage.

Used later for:

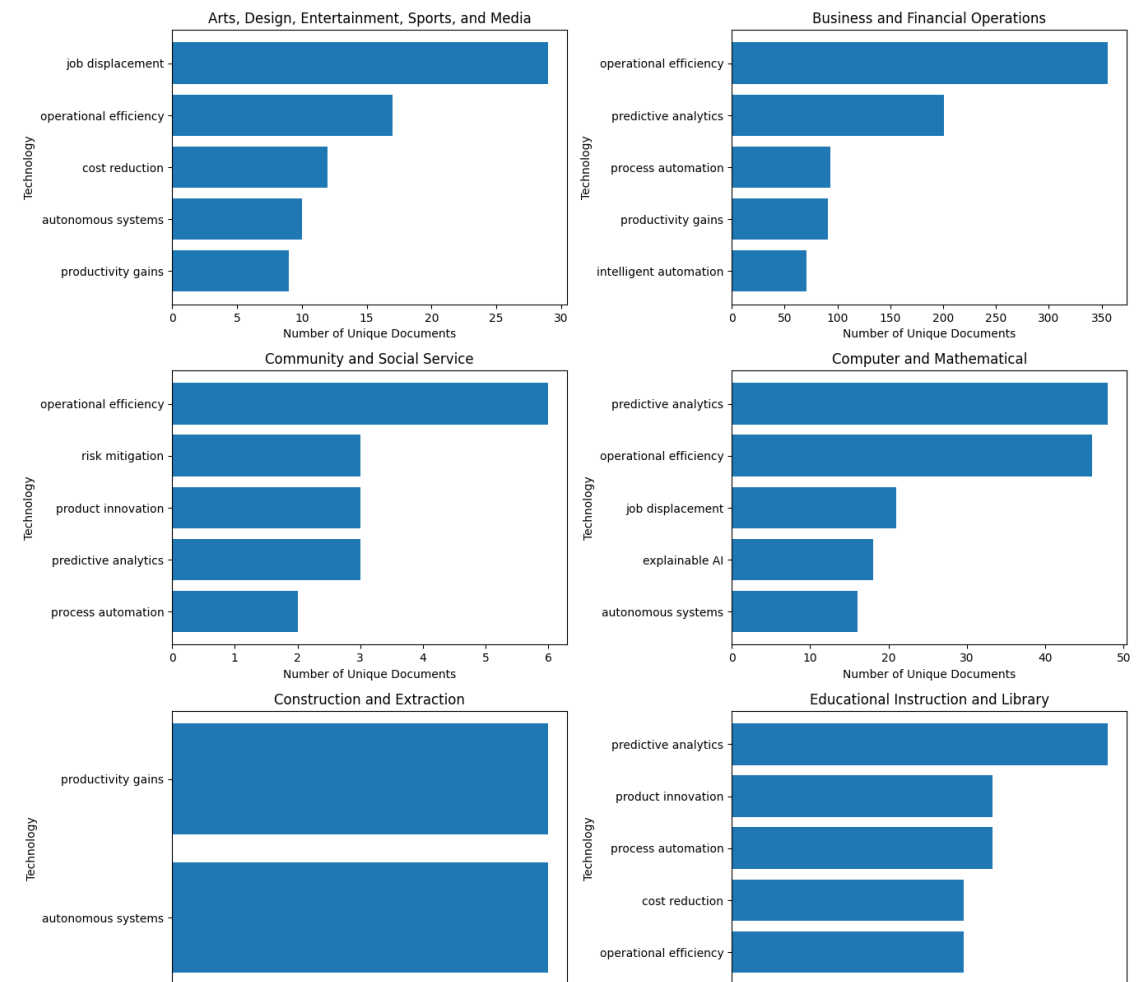
- Company-level sentiment profiles
- Industry × company hotspots
- Over-time sentiment trends

Most common AI “means” themes in the corpus

- **Arts, Design, Entertainment, Sports, and Media**
 - **Job displacement** dominates, suggesting stronger concern about creative labor substitution.
 - AI is also framed around **efficiency and cost reduction**, but the labor-risk narrative is most visible.
- **Business and Financial Operations**
 - AI is mainly discussed as a tool for **operational efficiency and predictive analytics**.
 - The sector shows a strong **automation and productivity** narrative.
- **Community and Social Service**
 - AI is framed more around **operational support and risk mitigation** than worker replacement.
 - Mentions of **product innovation** suggest a service-improvement angle.
- **Computer and Mathematical**
 - **Predictive analytics** and **operational efficiency** lead, reflecting data-heavy AI use cases.
 - **Job displacement** and **explainable AI** show that labor and governance concerns are also important.
- **Construction and Extraction**
 - The dominant themes are **productivity gains** and **autonomous systems**.
 - AI impact here appears tied to **physical operations and site automation**.
- **Educational Instruction and Library**
 - **Predictive analytics** leads, suggesting interest in data-driven educational support.
 - AI is also associated with **innovation, process automation, and cost reduction**.

Technology mix varies by industry (examples)

Top Technologies Mentioned Within Each Industry



Restriction: sentiment model is self-trained (fine-tuned)

Weak supervision pipeline

- Build entity-centered context windows around organization mentions
- Generate weak labels with an LLM (positive / neutral / negative / unclear)
 - Used GPT 5-nano, prompt:

```
prompt = (
    "You are a helpful and precise assistant for analyzing the sentiment of sentence(s) extracted from AI-related News articles.\n"
    "Your job is to create a sentiment label (tone) to this sentence. Especially the attitude towards the impact of AI on the industry and organization.\n"
    "The organization in the sentence is marked with <ENT> and </ENT> tags. Focus on the sentiment towards the entity.\n"
    "Rules:\n"
    "- Judge overall tone of the coverage.\n"
    "- If the article is mostly factual or mixed without a clear tilt, choose neutral.\n"
    "- If sentiment cannot be inferred, choose unclear.\n"
    "- Output exactly ONE word from: positive, neutral, negative, unclear.\n"
    "- No punctuation. No explanation.\n\n"
    f"Text:\n{text}"
)
```

- Fine-tune roberta-base on the weakly labeled dataset
- Validate on held-out set and inspect confusion matrix

Validation metrics (10k examples)

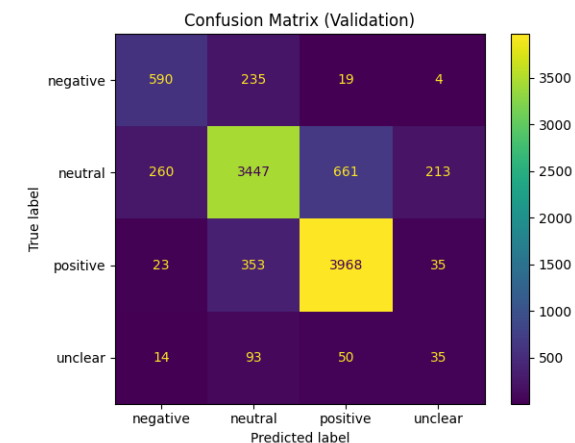
Accuracy: 0.804

Per-class F1:

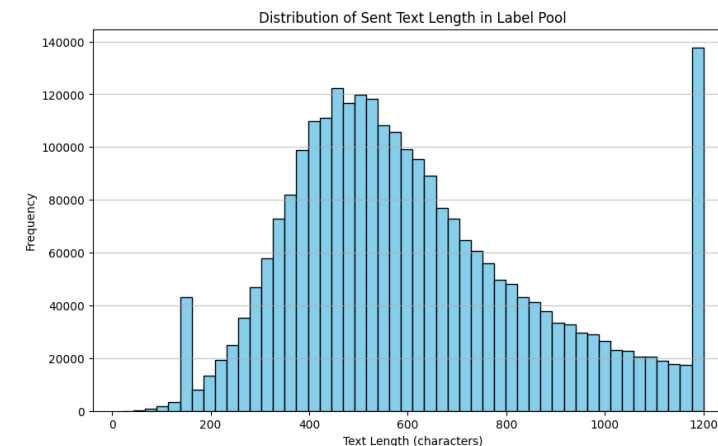
- positive: 0.874
- neutral: 0.792
- negative: 0.680
- unclear: 0.146

Note: "unclear" is rare and hardest to learn.

Confusion matrix (validation)



Context length distribution (label pool)



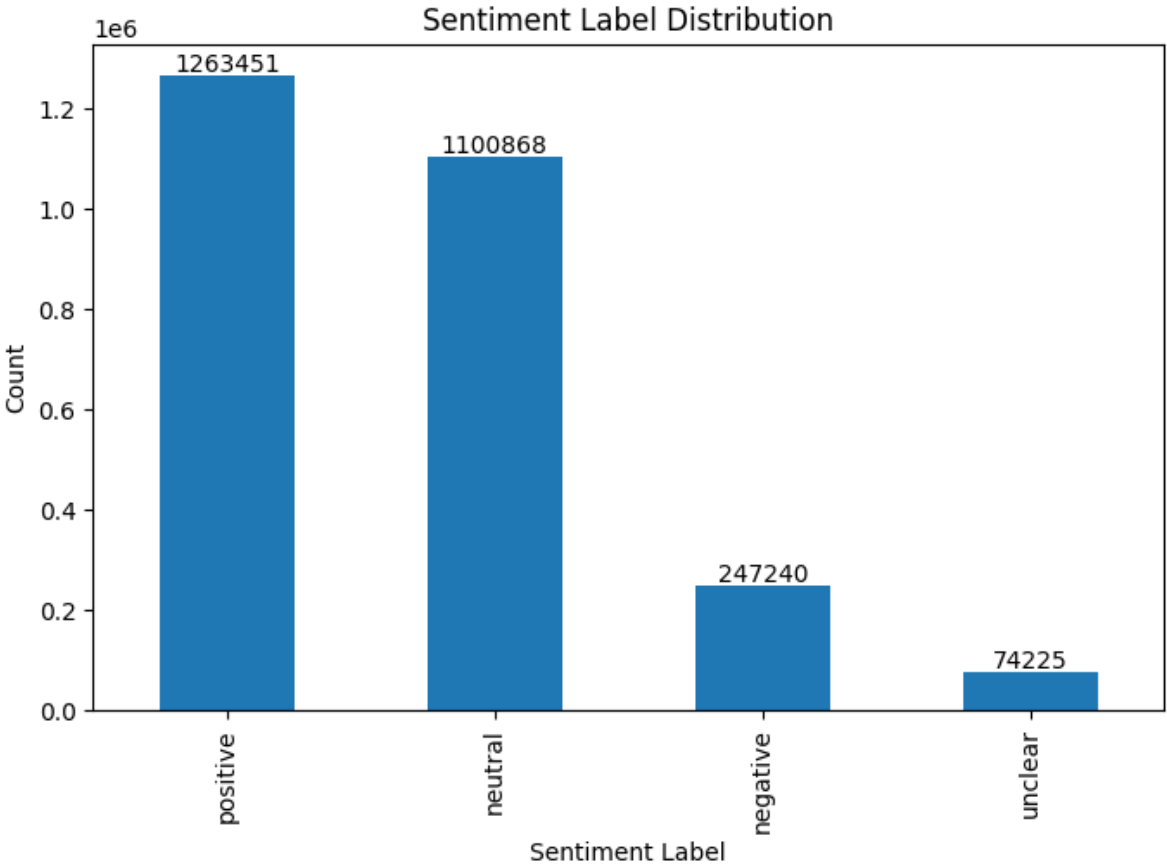
Apply the fine-tuned model to all entity-centered contexts, then aggregate by industry/company/tech/time

Total contexts labeled
2,685,784

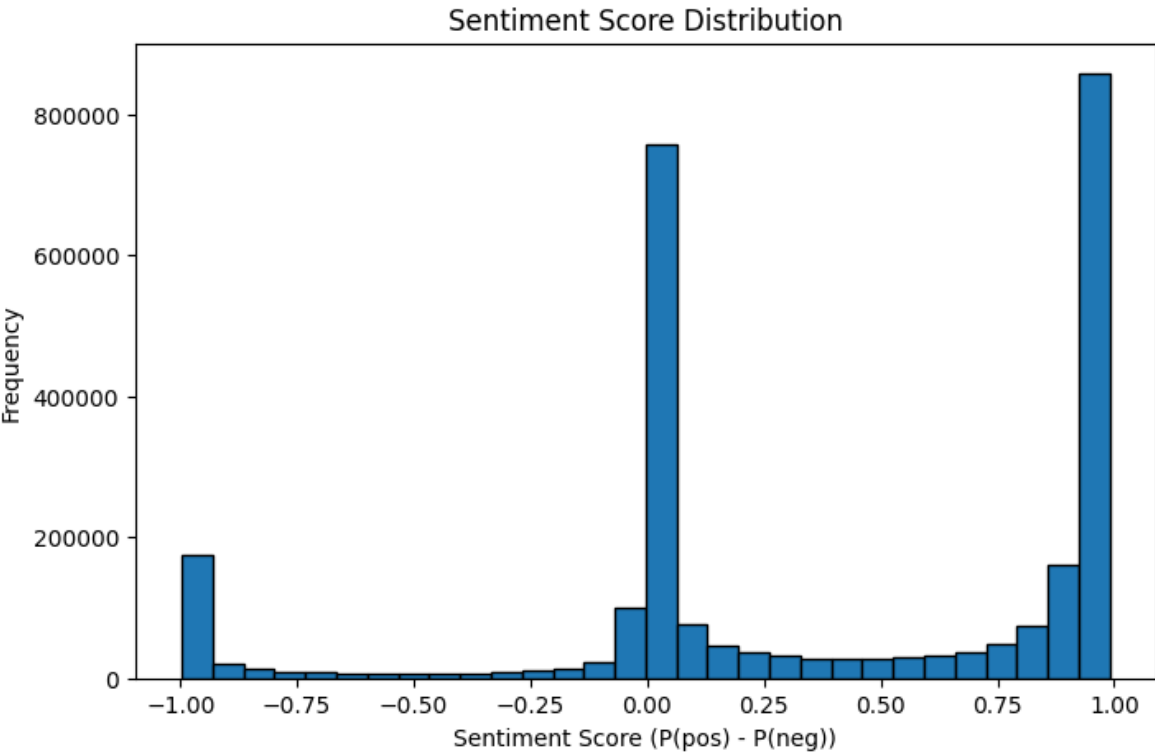
Label set
positive · neutral · negative · unclear

Score
 $P(\text{pos}) - P(\text{neg}) \in [-1, 1]$

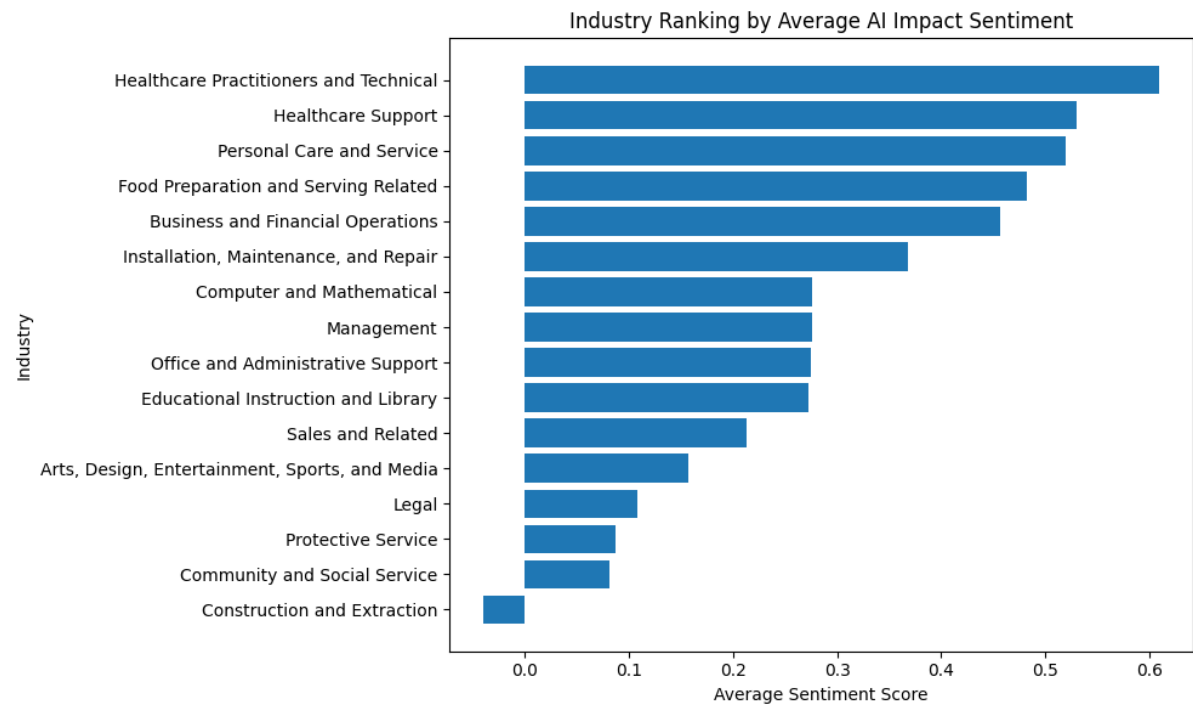
Label distribution (counts)



Score distribution



Average sentiment score by industry (higher = more positive framing)

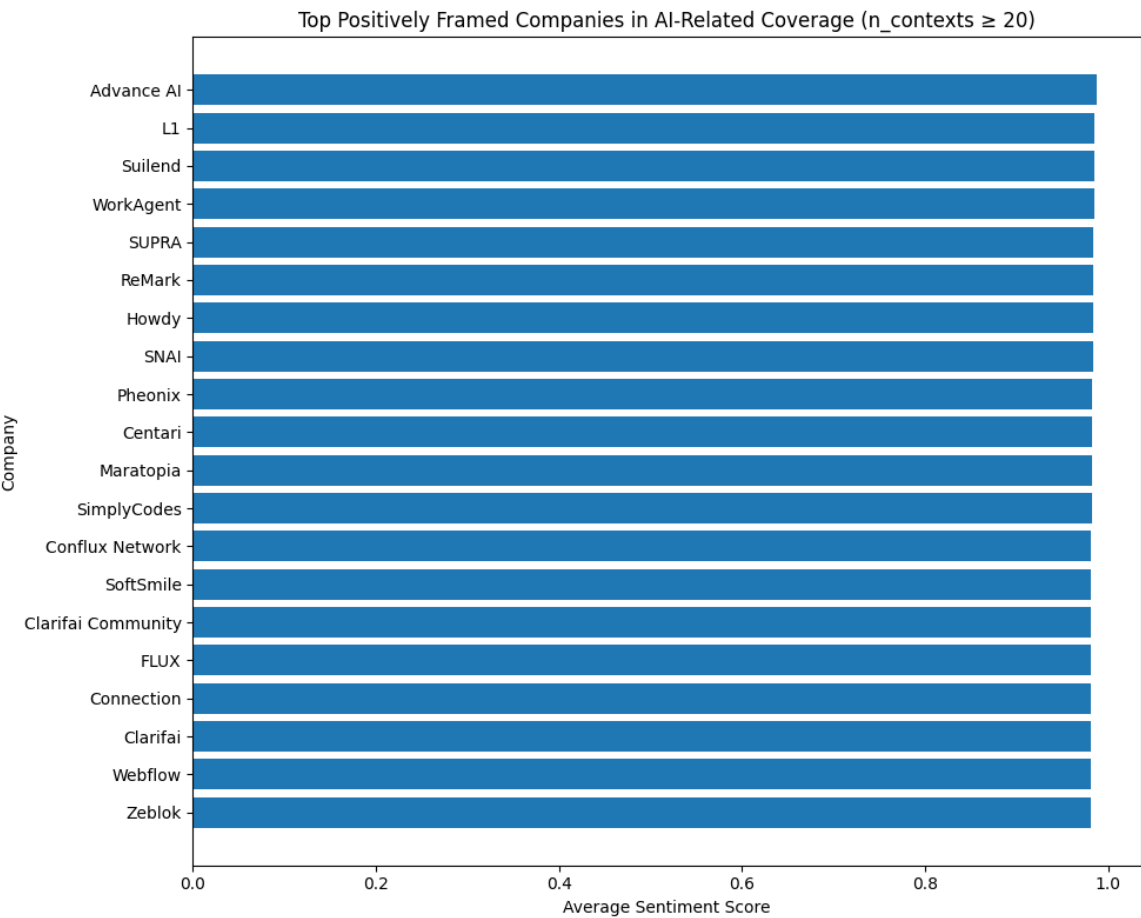


Sentiment composition (selected industries)



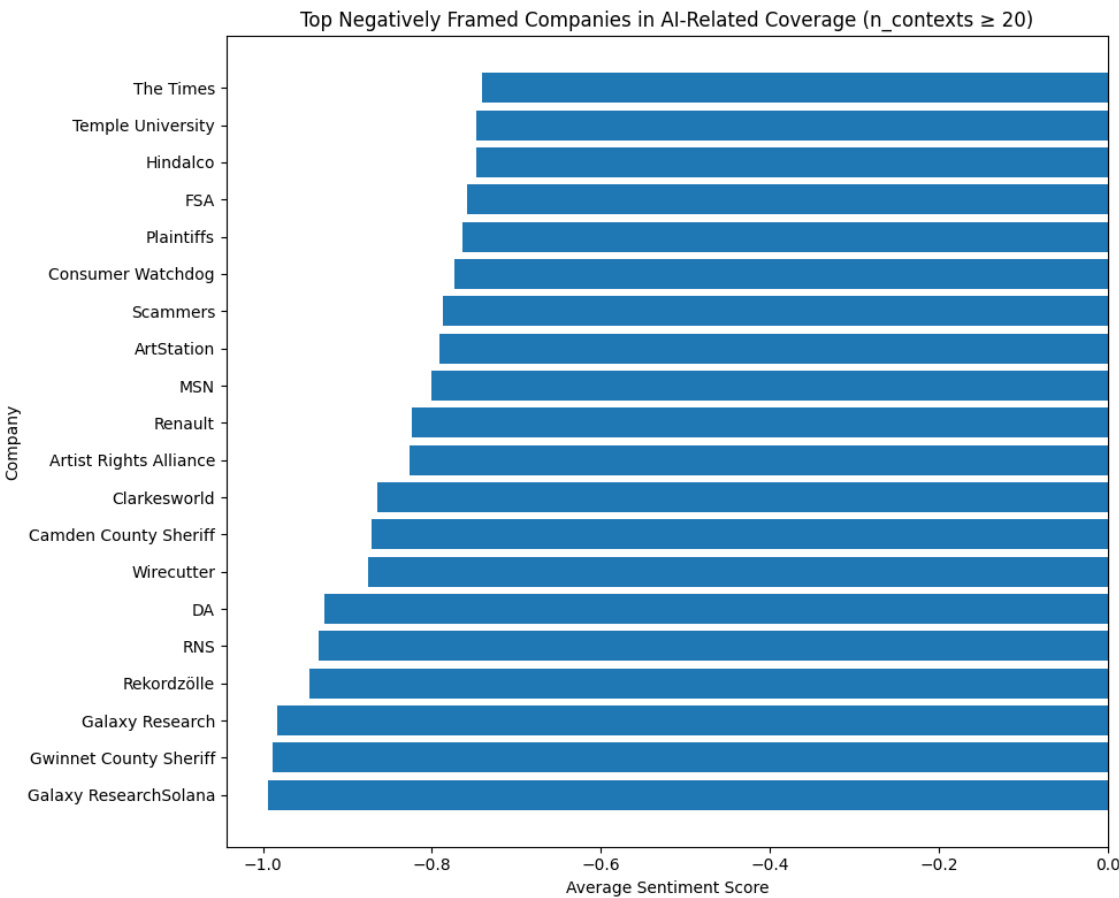
Company sentiment is computed by averaging sentiment scores across entity-centered contexts (min contexts ≥ 20)

Most positively framed (examples)

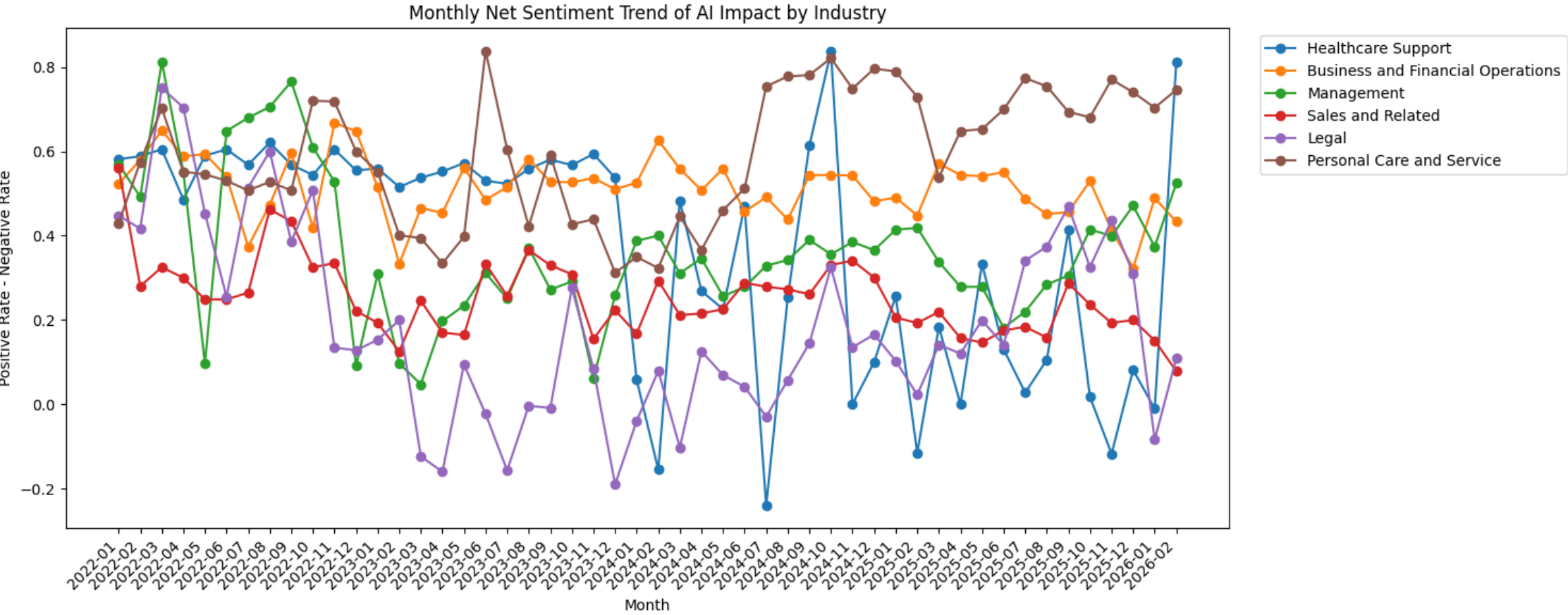


Caveat: entity canonicalization is imperfect (some names are publishers/generic nouns).

Most negatively framed (examples)



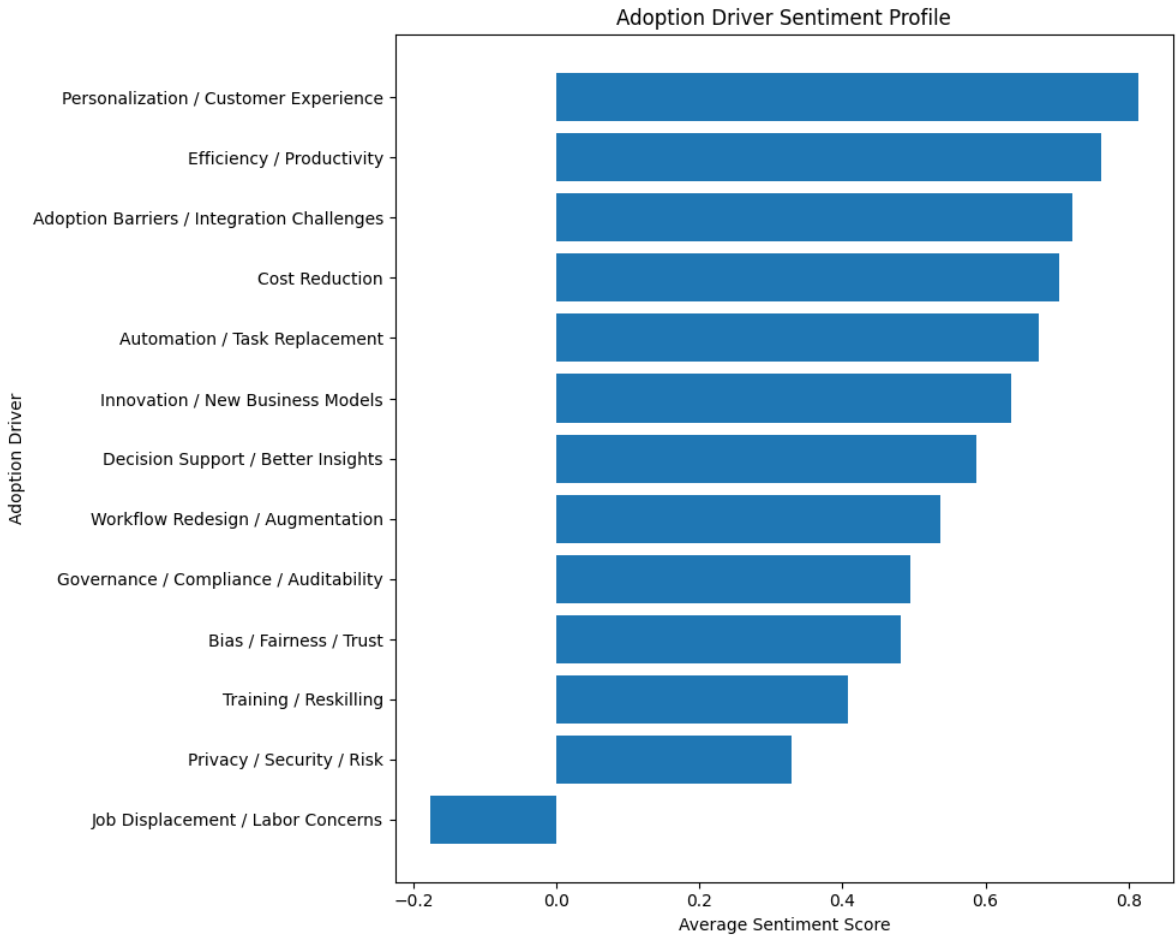
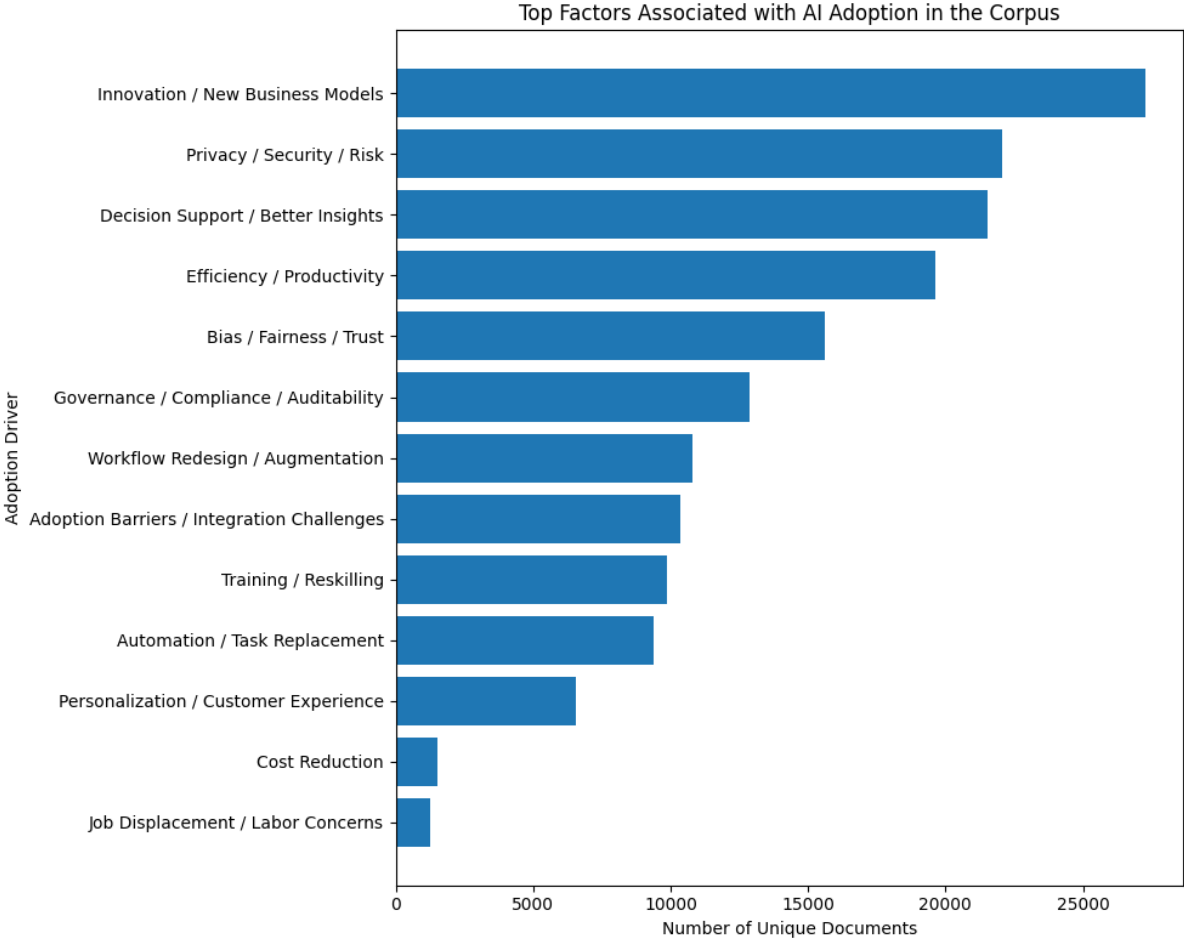
Net sentiment = positive rate – negative rate (by month, by industry)



Interpretation: compare trend direction (momentum) vs baseline level; spikes often align with major product/regulation news cycles.

Q3: What drives successful vs unsuccessful adoption?

We treat “adoption driver” as a technology/means theme associated with adoption contexts and analyze volume + sentiment

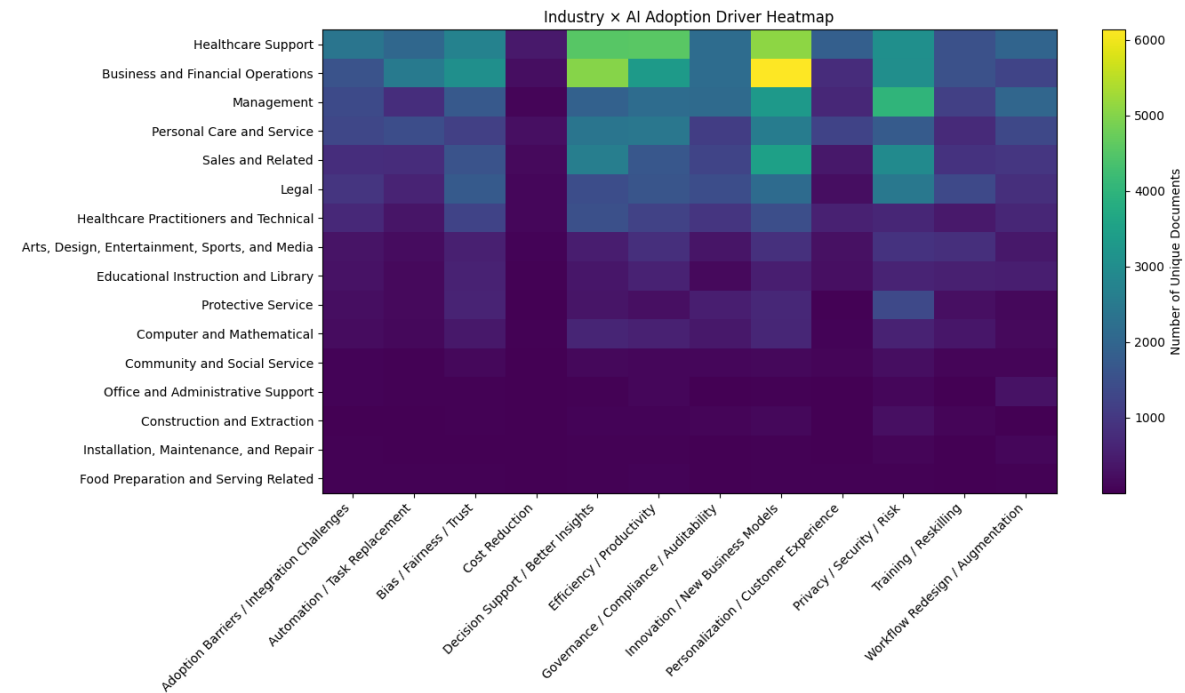
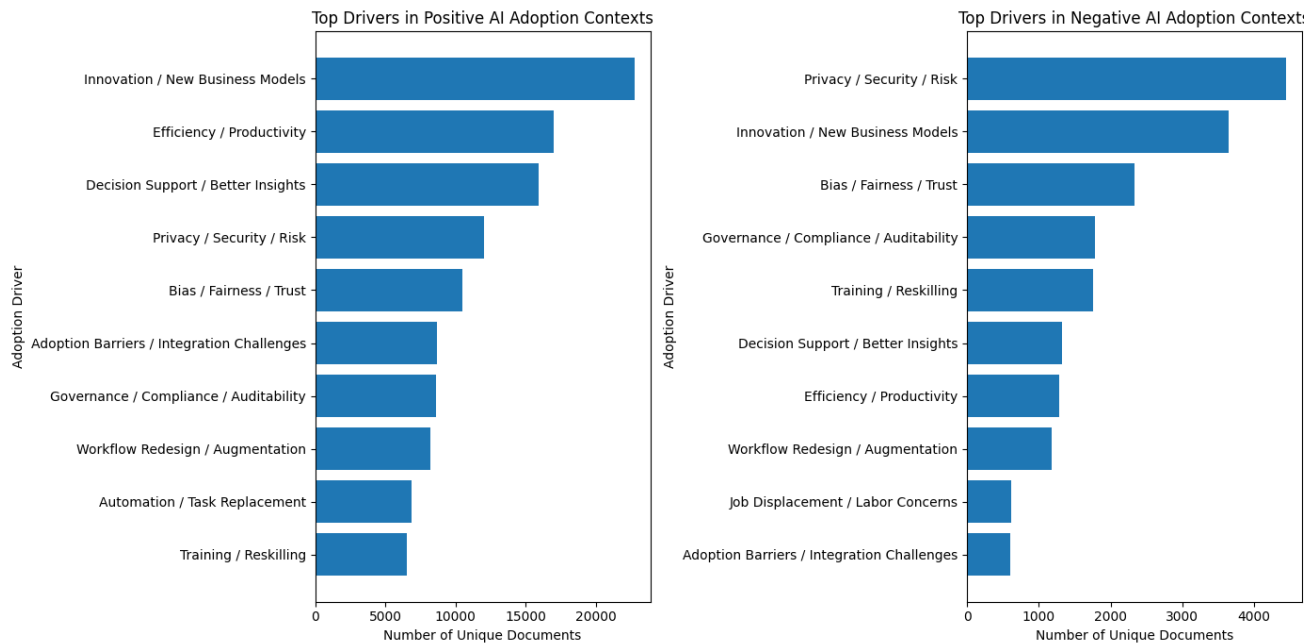


Key pattern High-volume drivers are mostly positive (innovation, productivity, decision support). Risk-heavy drivers (privacy/security, fairness, governance) reduce sentiment; “job displacement” is the only strongly negative driver.

Adoption drivers: what differs in negative contexts?

Drivers × Industry

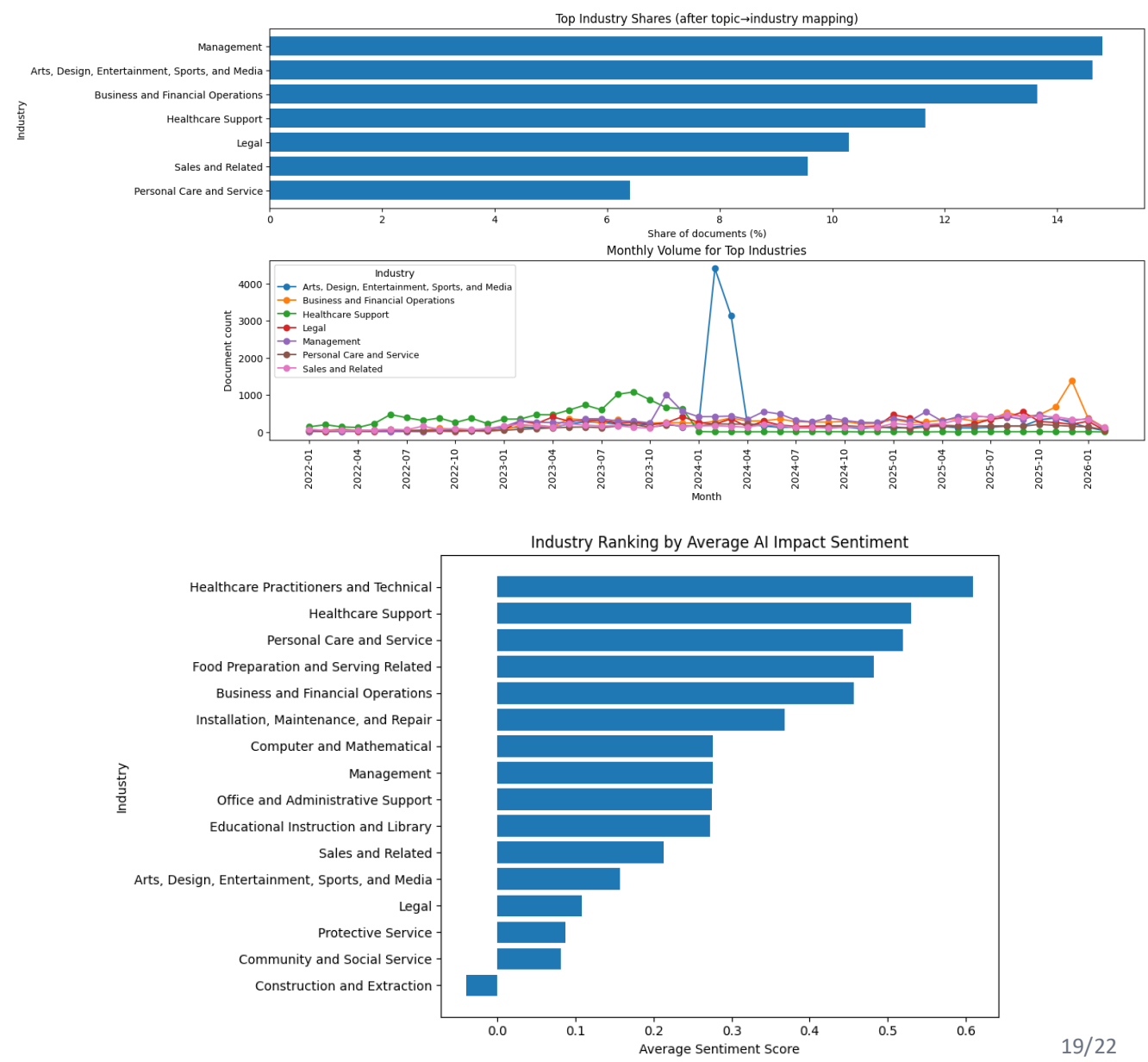
Compare drivers in positive vs negative adoption contexts; then view industry × driver co-occurrence



Wrap-up: Q1. Which industries are most likely to be impacted by AI?

Most exposed industries

- Management, Arts/Media, Business & Financial Operations, Healthcare Support, Legal, and Sales have the highest exposure.
- These are the sectors most frequently discussed in AI-impact news, especially in information-heavy and service-oriented work.
- Topic→industry mapping turns article-level topics into an interpretable industry impact map.
- High exposure highlights where workflow redesign, labor implications, and competitive pressure are most visible.



Wrap-up: Q2. AI's impact on those industries and companies mostly positive or negative?

Most exposed industries

AI's impact on those industries and companies: mostly positive or negative?

Answer: mostly mixed, with a slight positive tilt in some industries - but clearly not uniformly positive.

- **Industry-level sentiment is mixed rather than one-directional.**

The monthly net sentiment lines stay above zero for several industries much of the time, but the levels vary a lot across sectors and months.

Some industries are consistently more positive.

- **Business & Financial Operations and Personal Care & Service** show relatively high and stable net sentiment, suggesting AI is often framed there in terms of efficiency, service improvement, or business value.

Others are more volatile or closer to neutral.

- **Healthcare Support, Management, Sales,** and especially **Legal** fluctuate much more, indicating that AI coverage in these sectors includes both optimism and concern.

- **Company-level sentiment is highly dispersed within each industry.**

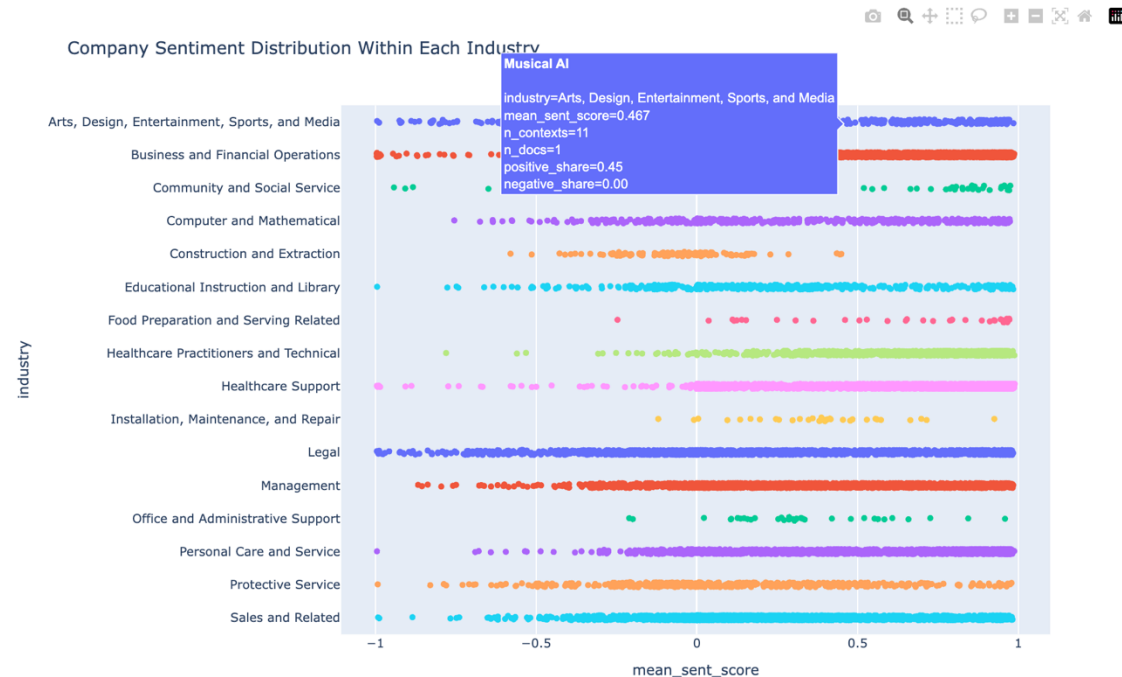
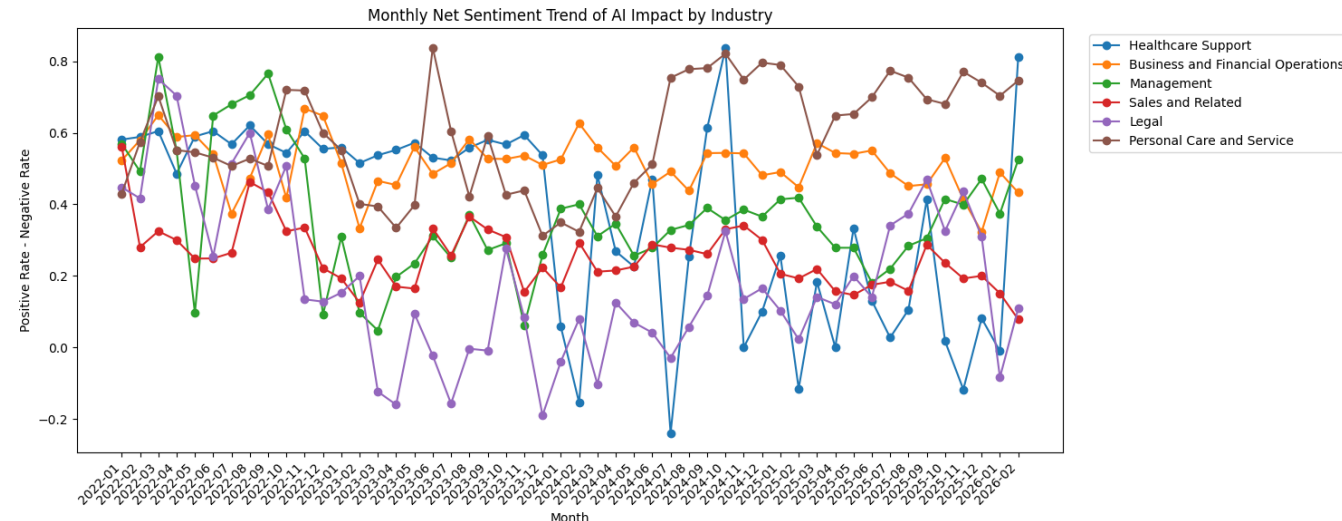
The company scatterplot shows organizations spread across the full range from negative to positive, which means firms in the same industry are not experiencing AI in the same way.

- **This suggests execution matters as much as industry.**

Industry gives the overall context, but company-level outcomes depend on how AI is adopted, governed, and integrated into workflows.

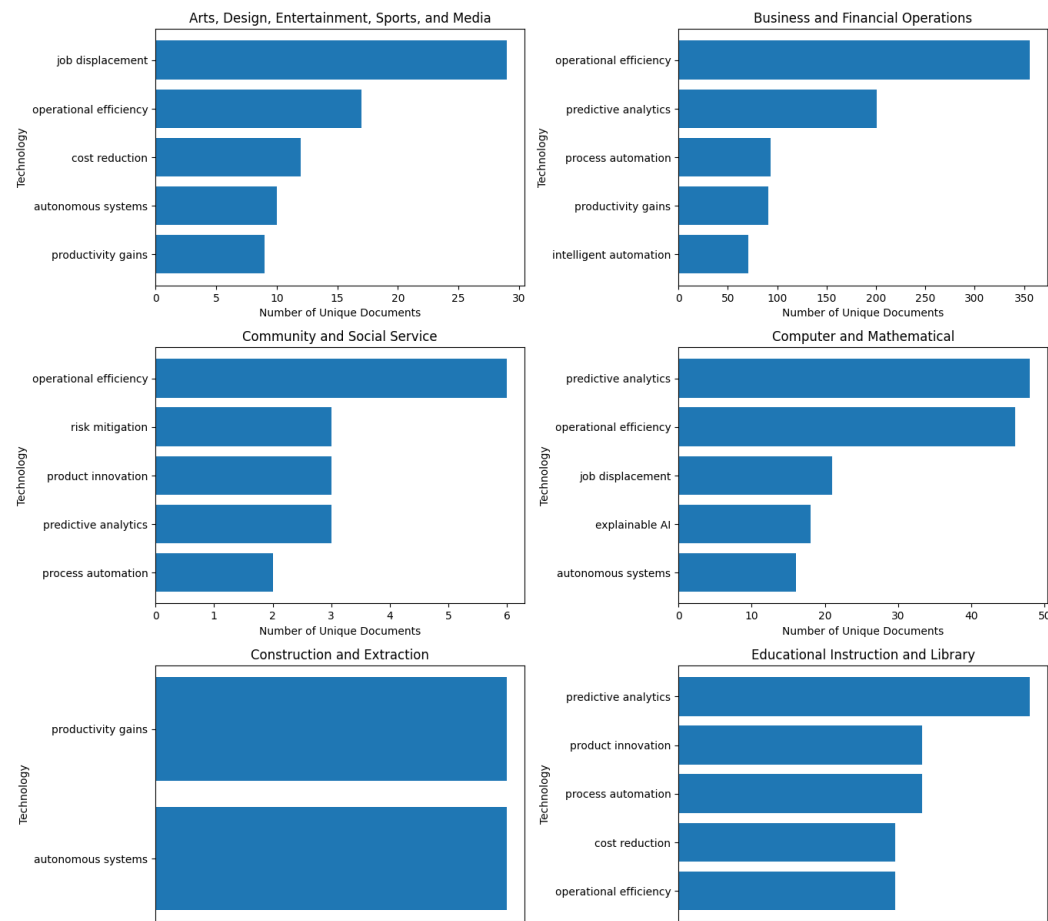
So the overall conclusion is: AI impact is neither mostly positive nor mostly negative.

It is **mixed**, with positive narratives around productivity and innovation, but also negative or uncertain narratives around displacement, risk, and implementation challenges.



- The most common mechanisms are predictive analytics, operational efficiency, automation, productivity gains, and autonomous systems.
- The dominant mechanism depends on the industry: job displacement is salient in Arts/Media, while efficiency and analytics dominate Business, Education, and Healthcare-related fields.
- AI is changing work through both task automation and process redesign—not through a single universal pathway.
- Successful adoption is more likely when AI is tied to clear use cases, operational fit, and measurable value.

Top Technologies Mentioned Within Each Industry



Final answers

Q1 · Who is impacted?

- High-exposure industries: Management, Arts/Media, Business & Finance, Healthcare Support, Legal, Sales.
- Most discussed entities include major AI/platform players (OpenAI, Google, Microsoft, Nvidia), plus industry-specific actors.

Q2 · How and by what means?

- Industry sentiment is mostly positive/neutral; negative share is higher in Legal and job-displacement contexts.
- Top means: operational efficiency, predictive analytics, process automation, productivity gains, AI copilots.

Q3 · What helps/hurts adoption?

- Success narrative: innovation/new business models, productivity, decision support, better customer experience.
- Failure narrative: privacy/security, bias/fairness/trust, governance/compliance, integration barriers.

Limitations & next steps

- Entity canonicalization: merge publisher/generic terms; strengthen org linking (aliases, KB).
- Topic outliers (-1): refine clustering hyperparameters and/or increase representation quality.
- Weak labels: spot-check a human-labeled subset for calibration; improve “unclear” class via targeted sampling.
- Causal interpretation: news sentiment is perception, not ground truth impact (use as leading indicator).